Scenario Summary

Wi-Fi Mesh Hotspot (meshConfig 0)

- Field: 400 × 400 × 30 m
- Mesh: 3×3 HWMP grid, spacing 200 m, AP height 1.5 m
- Buildings: Hybrid layout, cluster blocks shifted north
- Hotspot: 10 STAs (node 8 hosts two), IP 192.168.1.0/24
- Traffic window: 5–9 s (mesh settle 0–5 s)

STA	Traffic	Proto	Tx	Rx	PDR %	Delay ms	Throughput Mbps
0	HTTP small	TCP	3	3	100.00	2.18	0.00031
1	HTTP small	TCP	11	11	100.00	7.89	2.81
2	HTTPS large	TCP	866	758	87.57	71.45	2.22
3	HTTPS large	TCP	47	47	100.00	115.50	0.14
4	Video large	TCP	782	760	97.19	89.99	4.47
5	Video large	TCP	231	207	89.61	40.24	0.51
6	HTTP small	TCP	3	2	66.67	0.30	0.00024
7	VoIP small	UDP	8	3	37.50	2009.85	0.016
8	HTTP small	TCP	11	11	100.00	215.89	0.039
9	HTTP large	TCP	78	39	50.00	144.09	0.185

5G NR Playfield

- Field: 400 × 400 × 30 m urban macro
- gNBs: (-100,200,30) and (500,200,30)
- UEs: 10, Gauss–Markov 0.3–0.8 m/s
- Traffic: HTTP/HTTPS 2 MB, Video 3 MB, VoIP UDP 1.5 MB, HTTP 2.5 MB, Mixed 2 MB

UE	Traffic	Proto	PDR %	Delay ms	Throughput Mbps
0	HTTP	TCP	94.48	40.57	0.88
1	HTTP	TCP	94.15	21.97	1.46
2	HTTPS	TCP	94.14	17.75	1.45
3	HTTPS	TCP	94.08	25.92	1.36
4	Video	TCP	94.51	17.68	2.19
5	Video	TCP	96.05	11.72	2.76
6	VoIP	UDP	99.82	13.54	1.35
7	VoIP	UDP	96.85	37.73	1.29
8	HTTP	TCP	0.00	0.00	0.00
9	Mixed	TCP	94.34	17.18	1.83

LTE Playfield

- Field: 400 × 400 × 30 m urban macro

- eNBs: dual macro sites aligned with NR scenario
- Traffic: identical to NR (HTTP/HTTPS/Video TCP, VoIP UDP, Mixed TCP)

UE	Traffic	Proto	PDR %	Delay ms	Throughput Mbps
0	HTTP	TCP	94.11	88.69	0.55
1	HTTP	TCP	93.44	147.41	0.37
2	HTTPS	TCP	84.48	28.34	0.12
3	HTTPS	TCP	95.94	73.16	0.71
4	Video	TCP	93.29	91.20	0.54
5	Video	TCP	95.58	71.96	0.70
6	VoIP	UDP	16.28	322.98	0.22
7	VoIP	UDP	50.95	105.77	0.68
8	HTTP	TCP	76.13	53.46	0.08
9	Mixed	TCP	94.29	99.16	0.49